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| Project Title   | CUSTOMER LIFETIME VALUE (CLV) PREDICTION USING THE BG/NBD MODEL |
| Week            | Week 1                                                          |

## WEEK-1 TECHINCAL REPORT

### TASK NAME:

CUSTOMER LIFETIME VALUE (CLV) PREDICTION USING THE BG/NBD MODEL

### ABSTRACT:

Customer Lifetime Value (CLV) is an important business metric that estimates the total value a customer contributes throughout their relationship with a company. The Beta-Geometric/Negative Binomial Distribution (BG/NBD) model is a probabilistic model designed for non-contractual businesses where customers can stop purchasing without notifying the company. This project develops the BG/NBD model to predict future customer purchases and estimate CLV using historical transaction data. The model helps businesses identify valuable customers, improve retention strategies, and optimize marketing decisions.

### INTRODUCTION:

Businesses aim to retain profitable customers while minimizing marketing costs. Customer Lifetime Value provides an estimate of the future revenue expected from each customer. The BG/NBD model predicts purchasing behavior using transaction history without requiring subscription or cancellation information. It uses purchase frequency, recency, and observation time to estimate the probability that a customer is still active and the number of future purchases.

### THEORY:

The BG/NBD model is based on the following assumptions:

- Customer purchases follow a Poisson process.
- Purchase rates vary among customers according to a Gamma distribution.
- Customers may become inactive after any purchase.
- Dropout probabilities follow a Beta distribution.
- Customer activity is estimated from transaction history.

The model estimates four parameters:

- $r$  – Shape parameter

- **$\alpha$  (alpha)** – Scale parameter
- **$a$**  – Dropout shape parameter
- **$b$**  – Dropout scale parameter

These parameters are used to estimate customer activity and expected future purchases.

### **OBJECTIVES:**

- Predict future customer purchases.
- Estimate Customer Lifetime Value (CLV).
- Identify active and inactive customers.
- Support customer segmentation.
- Improve marketing and retention strategies.

### **METHODOLOGY:**

1. Collect customer transaction data.
2. Calculate Frequency, Recency, and Customer Age (T).
3. Estimate BG/NBD model parameters using Maximum Likelihood Estimation (MLE).
4. Predict future purchase frequency.
5. Calculate Customer Lifetime Value.
6. Analyze customer segments and business insights.

### **DATASET:**

The dataset contains:

- Customer ID
- Transaction Date
- Purchase Frequency
- Recency
- Observation Period (T)
- Purchase Amount (optional for CLV)

### **MATHEMATICAL MODEL:**

The BG/NBD model estimates:

$$\theta = (r, \alpha, a, b)$$

where:

- **r,  $\alpha$**  describe customer purchase behavior.
- **a, b** describe customer dropout behavior.

These parameters are estimated using Maximum Likelihood Estimation (MLE).

### COMPLEXITY ANALYSIS:

- Data preprocessing:  **$O(N)$**
- Likelihood computation:  **$O(N)$**
- Parameter optimization: Iterative optimization
- Memory complexity:  **$O(N)$**

where **N** is the number of customers.

### RESULTS:

The BG/NBD model successfully:

- Predicts future customer purchases.
- Estimates Customer Lifetime Value.
- Identifies high-value customers.
- Supports customer retention and personalized marketing.
- Improves business decision-making through customer analytics.

### APPLICATIONS:

- E-commerce
- Retail Business
- Banking
- Insurance
- Subscription Services
- Customer Relationship Management (CRM)
- Marketing Analytics
- Sales Forecasting

## **CONCLUSION:**

The BG/NBD model is an effective probabilistic approach for Customer Lifetime Value prediction in non-contractual businesses. By analyzing customer purchase history, it accurately predicts future buying behavior and customer activity. These insights help organizations improve retention strategies, optimize marketing campaigns, and maximize long-term profitability.

## **REFERENCES:**

1. Fader, P. S., Hardie, B. G. S., & Lee, K. L. (2005). Counting Your Customers the Easy Way: An Alternative to the Pareto/NBD Model.
2. Schmittlein, D. C., Morrison, D. G., & Colombo, R. (1987). Counting Your Customers: Who Are They and What Will They Do Next?
3. Research articles on BG/NBD and Customer Lifetime Value prediction.